AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT395

Applied Cyber Security

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Final Examination
Spring Semester 2025/2026

Date: 8/June/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 h

Instructions:

- Total Marks: 50
- Total Number of Questions:5
- Marks Distribution:
  - Question 1: 10
  - Question 2: 10
  - Question 3: 10
  - Question 4: 10
  - Question 5: 10
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted.
- Be sure you write your name and ID on the External Answer booklet.

Question 1: Answer the following questions [10 Marks]

1. Which of the following is a security control that authenticates and authorizes users based on their roles and responsibilities?
   a) ACL
   b) VPN
   c) IDS
   d) RBAC
2. Which of the following is a security principle that ensures data is not altered or tampered with during transmission?
   a) Confidentiality
   b) Integrity
   c) Availability
   d) Non-repudiation
3. Which security system detects intrusions by comparing abnormal behavior with normal system activity?
   a) Host Intrusion Detection System (HIDS)
   b) Packet filter
   c) Internet Relay Chat (IRC)
   d) Personal software firewall
4. Which of the following is a social engineering attack that involves manipulating individuals to disclose sensitive information?
   a) Phishing
   b) Spoofing
   c) DDoS
   d) Brute force
5. Which type of software secretly monitors a user's online activities, collects information without consent, and sends it to a third party?
   a) Malware
   b) Spyware
   c) Adware
   d) All of the above
6. Which wireless security standard offers better protection compared to WEP?
   a) WPA
   b) WEP2
   c) TKIP
   d) MAC filtering
7. In a one-way password file, how is the password stored?
   a) By jumbling the password before storing it.
   b) By XORing the password with a key.
   c) By storing the hash value of the password.
   d) By passing the password through a PRF and storing it.
8. Ian has been offered a role that mainly focuses on determining user security requirements. Which security position best fits this role?
   a) Security administrator
   b) Security technician
   c) Security officer
   d) Security manager
9. A _____ can be a hardware device or a software program that filters all the packets of data that come through a network, the internet, etc.
   a) Firewall
   b) Antivirus
   c) Malware
   d) Cookies
10. Before encryption, what type of text is given as input to a cipher algorithm?
    a) Scalar text
    b) Complex text
    c) Transformed text
    d) Plain text

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Question 2: Answer the following questions [10 Marks]

1. What is multi-factor authentication (MFA), and why is it important? (3 Marks)
2. What are some common types of cyber threats that organizations face? (3 Marks)
3. What are the main types of malwares, and how do they differ? (2 Marks)
4. What is the role of encryption in ensuring data security? (2 Marks)

Question 3: Answer the following questions [10 Marks]

1. List different strategies for reducing risk. (2 Marks)

2. List a four best practices for managing passwords to enhance security? (2 Marks)

3. What is the role of redundancy in achieving fault tolerance? (2 Marks)

4. What are the three essential elements of Business Continuity Planning (BCP)? (2 Marks)

5. What is the purpose of a Business Impact Analysis (BIA)? (2 Marks)

Question 4 Answer the following questions [10 Marks]

Case Study (1); A company stores all employee data on a shared server without encryption. Employees use weak passwords, and there is no multi-factor authentication. Recently, sensitive employee records were leaked online.

Questions. (4 mark)

1. Identify the vulnerabilities in this setup:

2. Explain how attackers likely bypassed security:

3. Which CIA components were violated?

4. Recommend a comprehensive security upgrade plan:

Case Study (2); An online shopping website does not use HTTPS, and user passwords are stored in plain text. Attackers intercepted customer data and gained access to accounts.

Questions. (4 mark)

1. Identify the vulnerabilities in this setup:

2. Explain how attackers likely bypassed security:

3. Which CIA components were violated?

4. Recommend a comprehensive security upgrade plan.

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Case Study (3); A bank allows unlimited login attempts without account lockout. There is no monitoring system in place. Attackers gained access to multiple accounts by trying many password combinations.

Questions. (2 mark)

1. Identify the vulnerabilities in this setup:

2. Explain how attackers likely bypassed security:

3. Which CIA components were violated?

4. Recommend a comprehensive security upgrade plan.

Question 4 Match the cybersecurity term with its corresponding definition: [10 Marks]

1. An attack where files are locked and payment is demanded to unlock them.	a) Malware
2. A private string of characters used to log into accounts.	b) VPN (Virtual Private Network)
3. A method that asks for more than one proof of identity to log in.	c) Worm
4. A hidden program that tracks user activity and steals information.	d) Password
5. A fake page that looks real to trick users into entering credentials.	e) Spyware
6. A stored copy of files used in case of data loss.	f) Firewall
7. A flaw in a system that can be used by attackers.	g) Multi-factor authentication (MFA)
8. A malicious program that spreads automatically across devices.	h) Ransomware
9. A system that watches network traffic to detect suspicious behavior.	i) Data backup
10. A secure tunnel used to protect data sent over the internet.	j) Intrusion Detection System (IDS) k) Security vulnerability l) Spoofed website

End of Questions

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